



Day 35



“CLOUD SECURITY”

Cloud Storage encryption tool:

1. Boxcryptor
2. Veracrypt
3. AxCrypt etc.

Tips to protect Encryption Keys:

1. Use long and strong encryption keys
2. Develop plans to handle problems
3. Restrict access to keys
4. Decentralize encryption and decryption
5. Use centralized key management
6. Maintain user profiles
7. Lifecycle management of encryption keys
8. Store encryption keys securely

Hashing:

1. Definition – Hashing is the process of converting input data into a fixed-length string (hash value) using algorithms.
2. Purpose – Ensures data integrity, verifies authenticity, and secures passwords without storing them directly.
3. Examples – Common algorithms include MD5, SHA-1, SHA-256; used in digital signatures, file verification, and authentication.

Data Masking:

1. Definition – Data masking hides or alters sensitive information so it looks real but is unusable to attackers.
2. Purpose – Protects confidential data (like PII, financial info) while allowing safe use in testing, training, or analytics.
3. Techniques – Methods include substitution, shuffling, encryption, and nulling out sensitive field

Tokenization:

1. Definition – Tokenization replaces sensitive data (e.g., credit card numbers) with a random token that has no exploitable value.
2. Purpose – Protects sensitive information while still allowing systems to process transactions using tokens.
3. Difference from Encryption – Unlike encryption, tokens cannot be reversed without the mapping system (token vault).

Anonymization:

1. Definition – Anonymization removes or modifies personal identifiers so data can no longer be traced back to an individual.
2. Purpose – Protects user privacy while allowing organizations to analyze data safely.
3. Techniques – Includes generalization, suppression, data swapping, and aggregation to make re-identification impossible.

--The End--